MILITARY SPECIFICATION SHEET

READOUT, INCANDESCENT TYPE, STYLE I
SEGMENTED, DOT MATRIX/SEGMENTED, BAR MATRIX
INCANDESCENT LAMP/ISOLATION DIODE, TYPE R06

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the readouts described herein shall consist of this specification and the latest issue of MIL-R-28803.

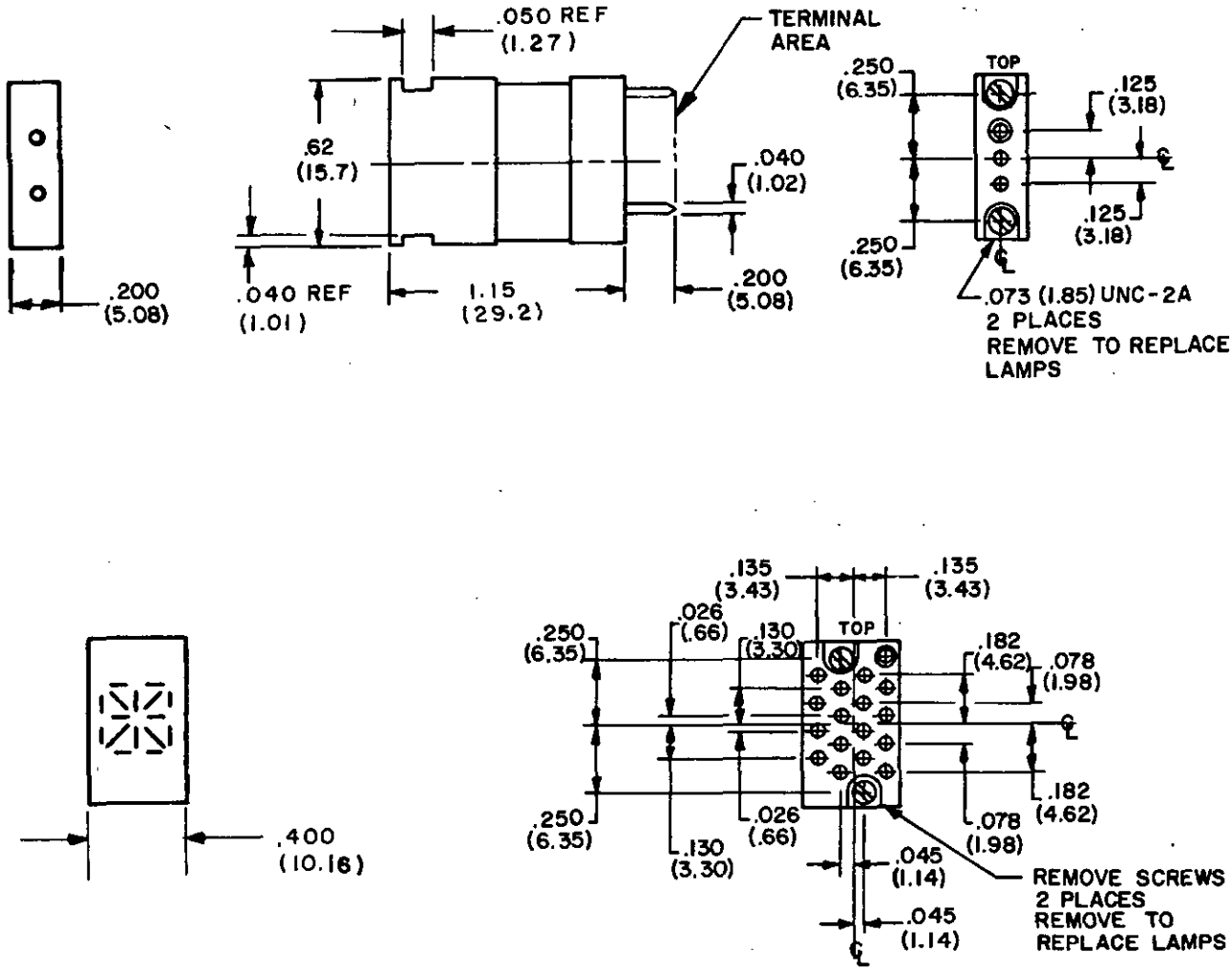
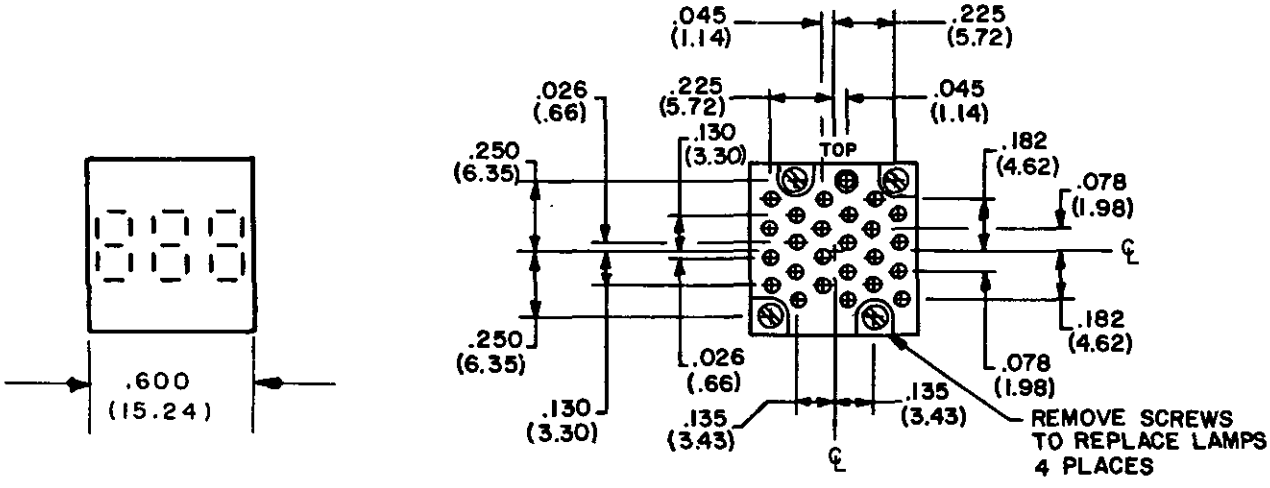


FIGURE 1. Type R06 readout module assemblies.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional with envelope as shown.

FIGURE 1. Type R06 readout module assemblies - Continued.

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TRUTH TABLE				
CIRCUIT CONFIG. I				
INPUT		DISPLAY	DOT	
A	B		A	B
○	○	COLON	△	△
	○	DECIMAL		△
	○	DEGREE		△

TABLE I. Truth table - 2 dot direct wire colon display.

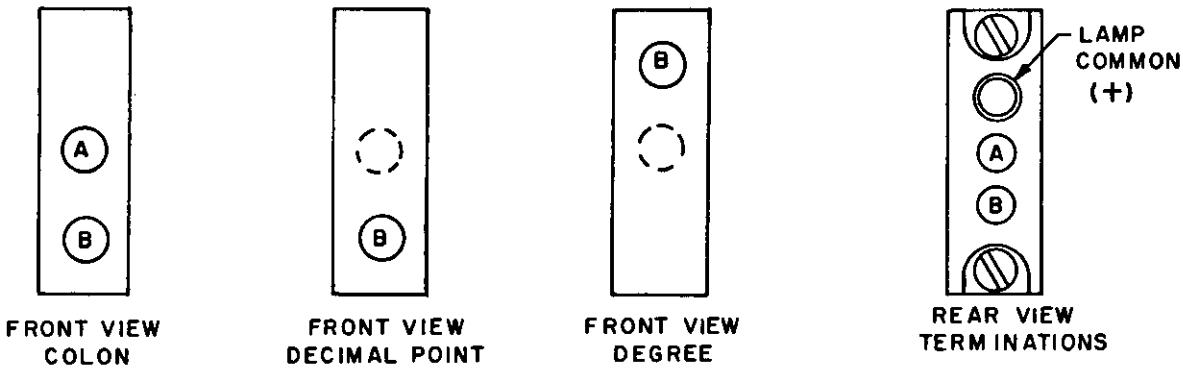


FIGURE 2. All colon display configurations.

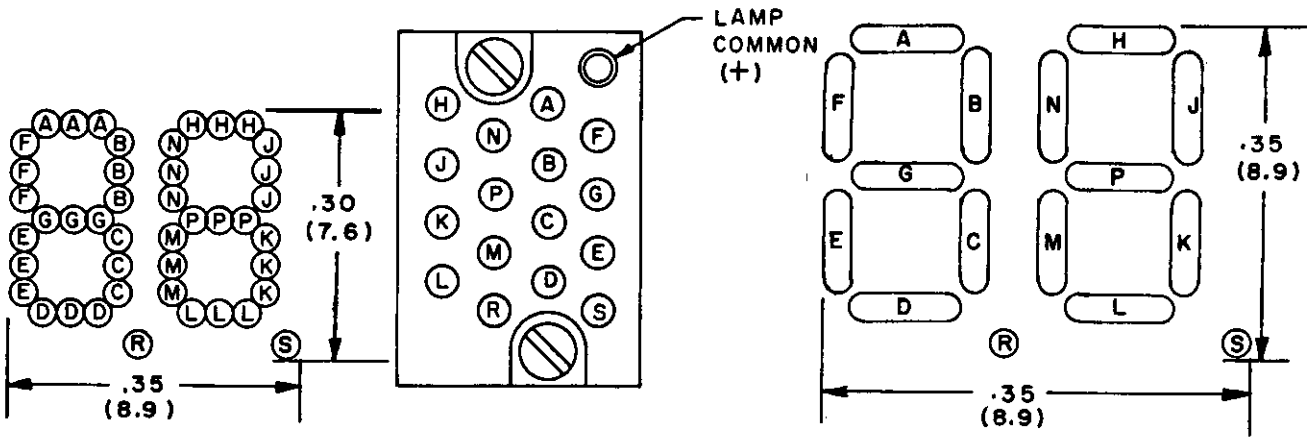


FIGURE 3. Segment terminations for tables II and III.

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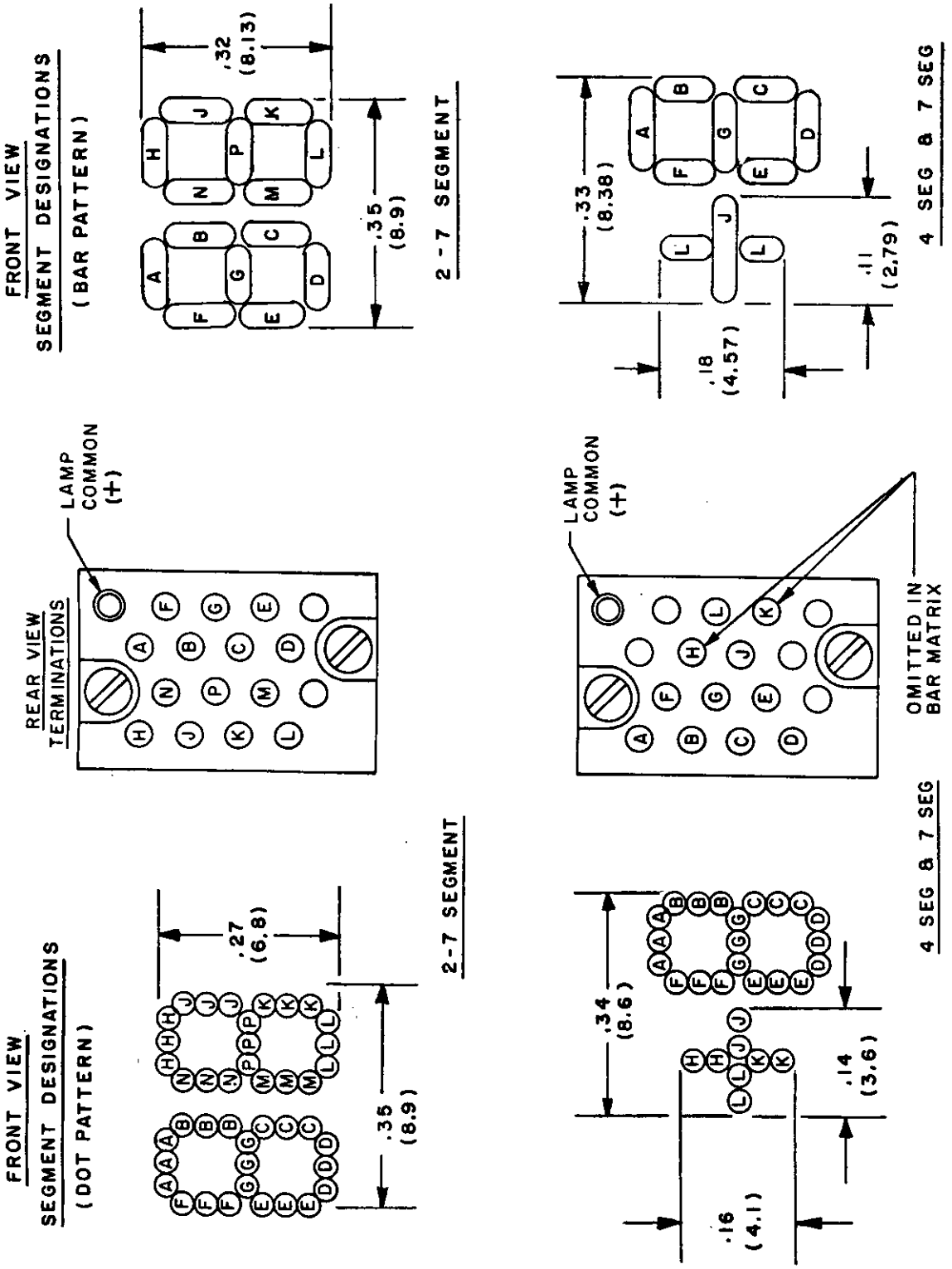


FIGURE 3. Segment terminations for tables II and III - Continued.

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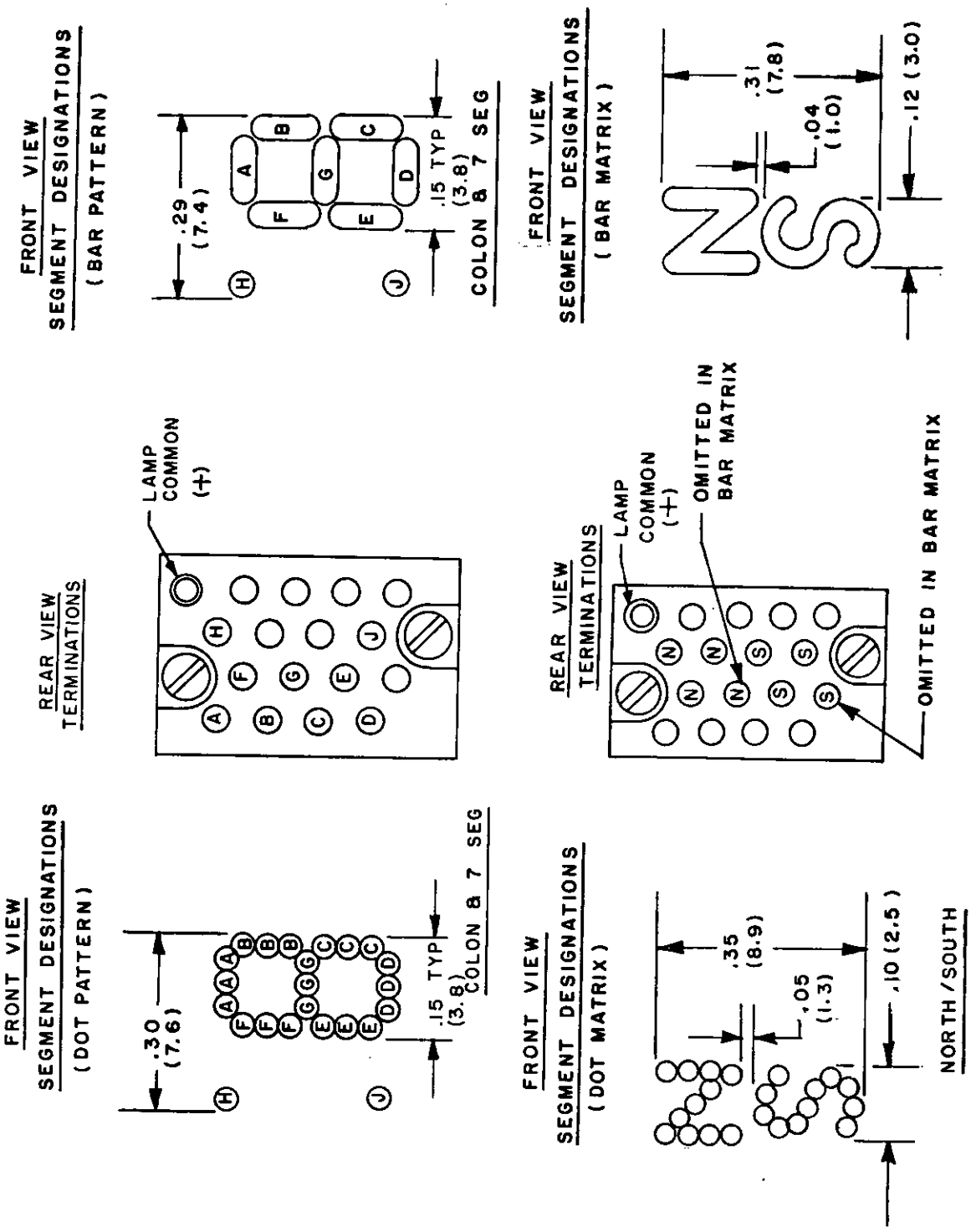


FIGURE 3. Segment terminations for tables II and III - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:00Z
Check the source to verify that this is the current version before use.

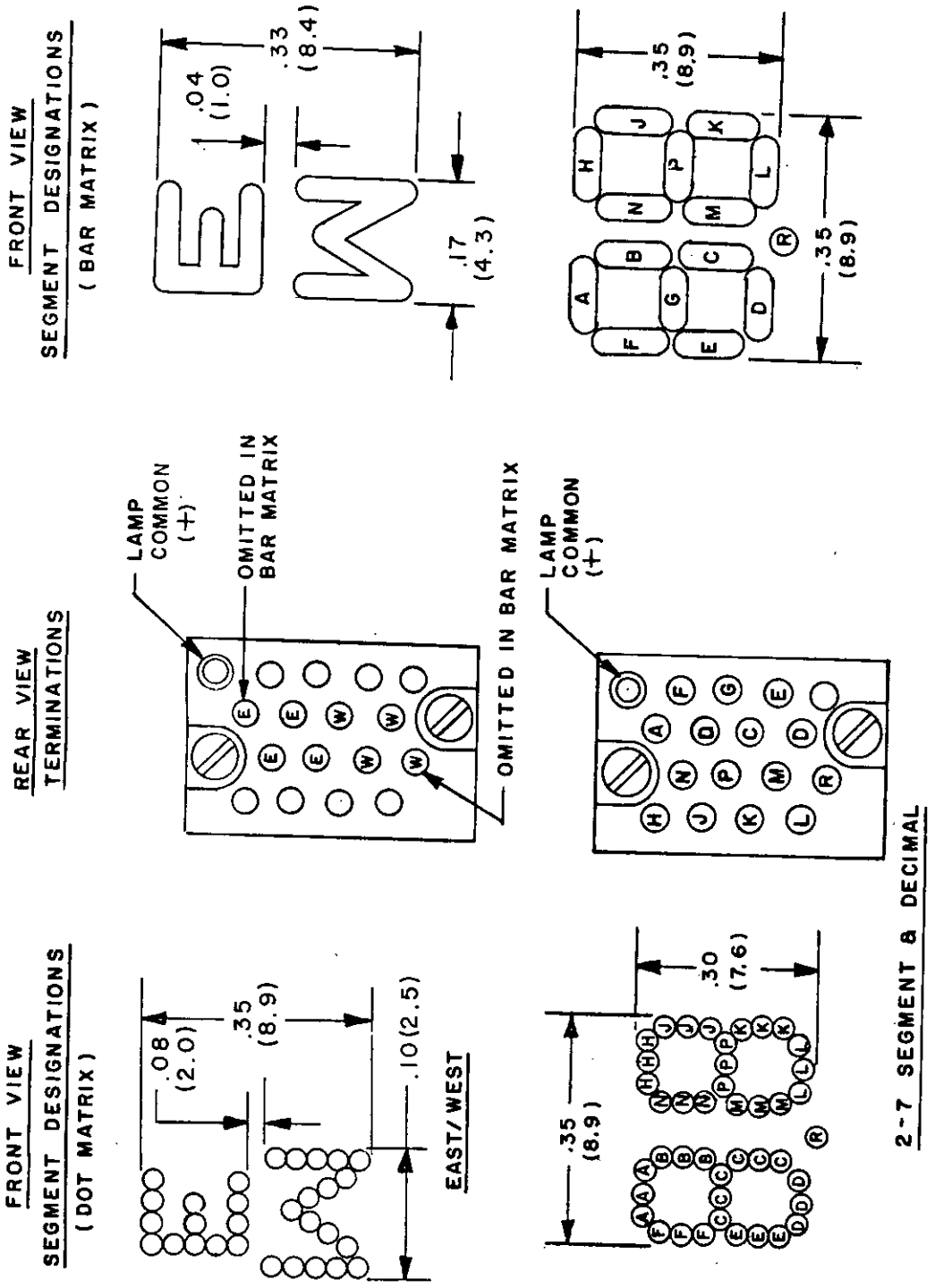


FIGURE 3. Segment terminations for tables II and III - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:00Z
Check the source to verify that this is the current version before use.

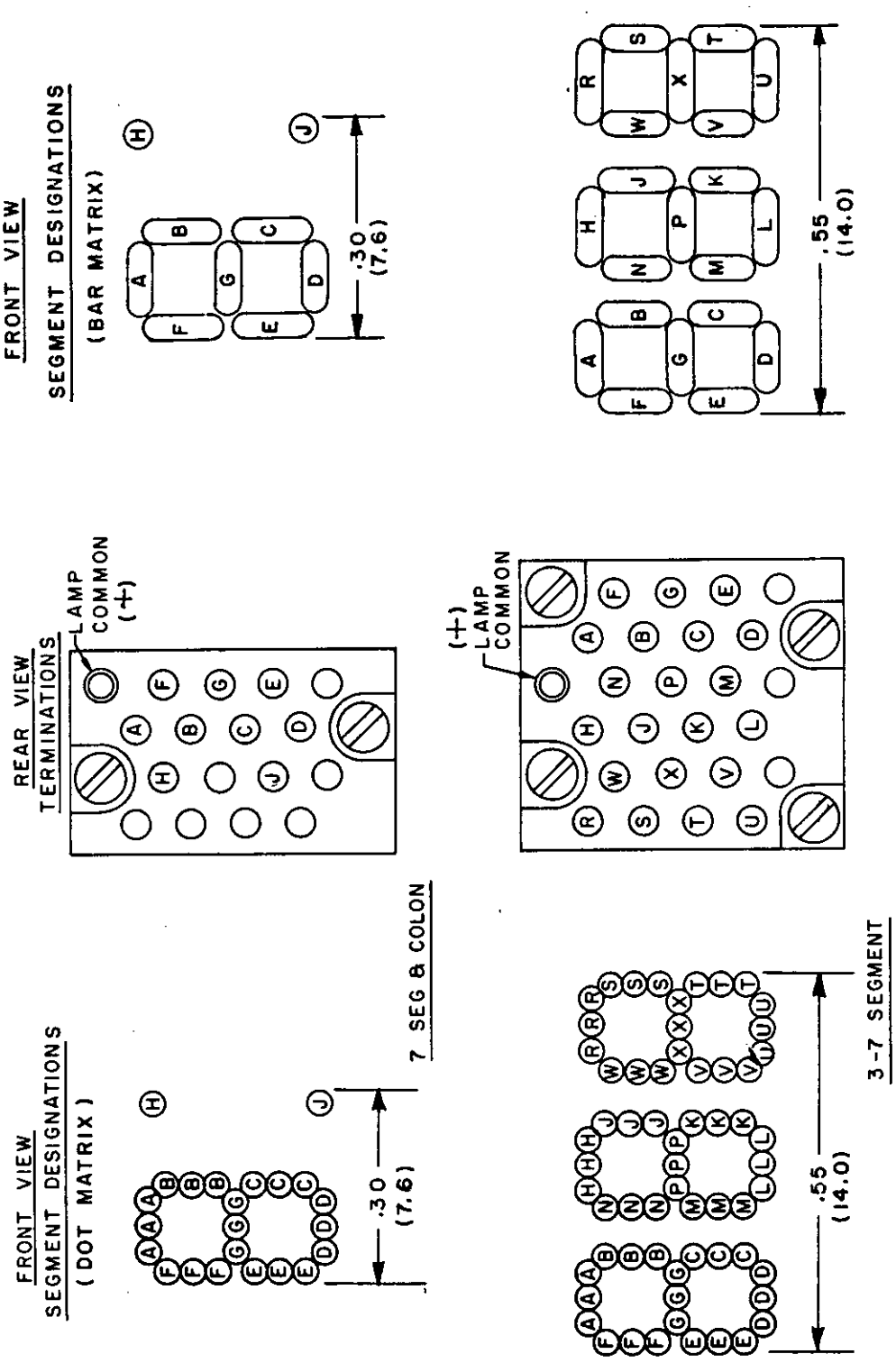


FIGURE 3. Segment terminations for tables II and III - Continued.

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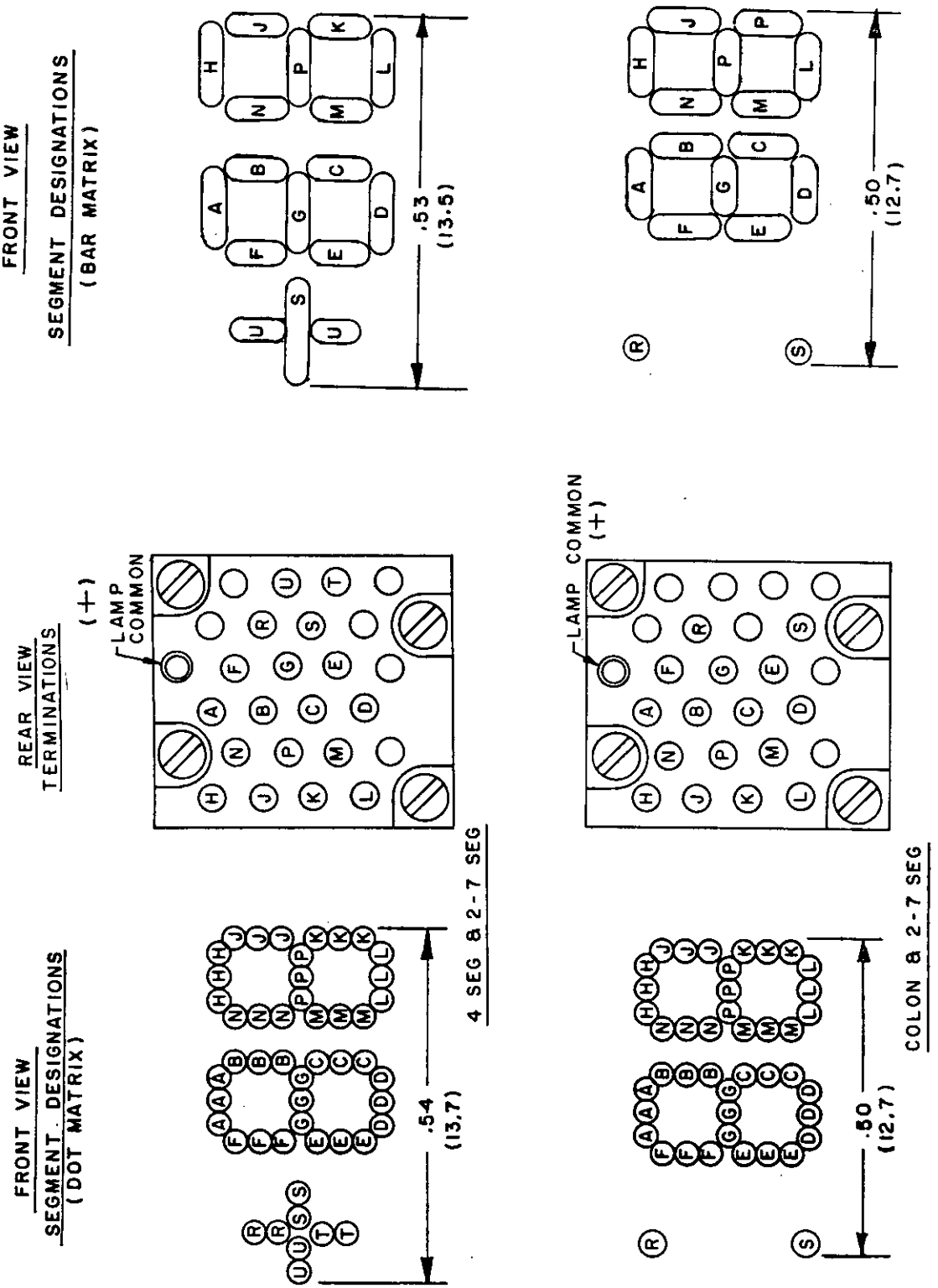


FIGURE 3. Segment terminations for tables II and III - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:00Z
Check the source to verify that this is the current version before use.

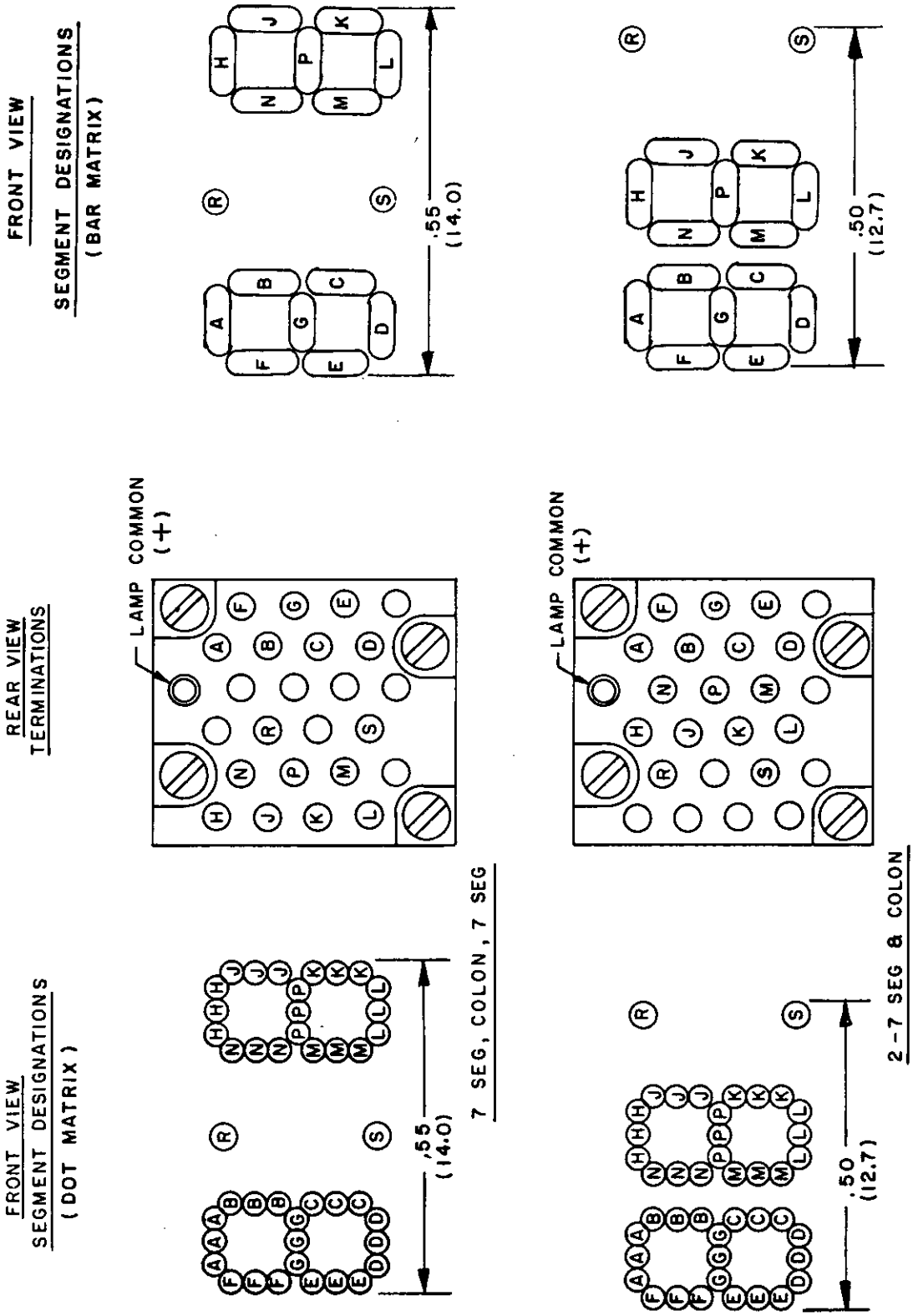
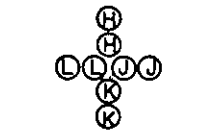


FIGURE 3. Segment terminations for tables II and III - Continued.

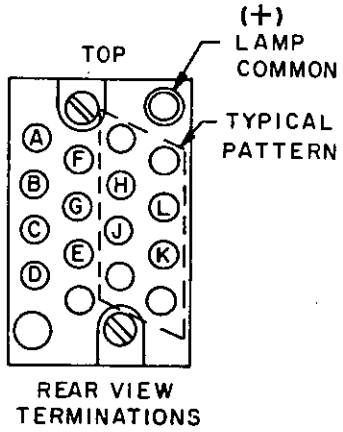
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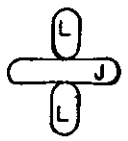
FRONT VIEW
(DOT MATRIX)

TRUTH TABLE								
CIRCUIT CONFIGURATION 2								
INPUT 1/				DISPLAY	SEGMENT LIT 1/			
H	J	K	L		H	J	K	L
○	○	○	○	+	△	△	△	△
	○		○	—		△		△

1/ Typical designators H thru J = R thru U.



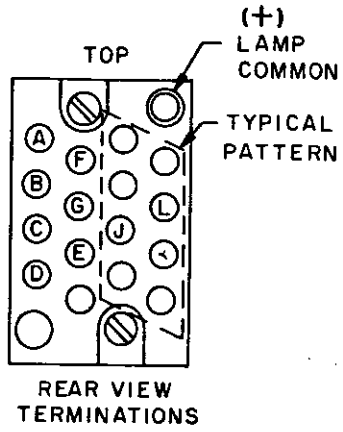
REAR VIEW
TERMINATIONS



FRONT VIEW
(BAR MATRIX)

TRUTH TABLE								
CIRCUIT CONFIGURATION 2								
INPUT 1/				DISPLAY	SEGMENT LIT 1/			
J	L				J	L		
○	○			+	△	△		
○				—	△			

1/ Typical designators J and L = S and U.



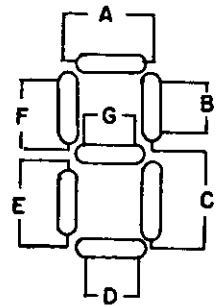
REAR VIEW
TERMINATIONS

TABLE II. Truth table for 4 segment direct wire sign display.

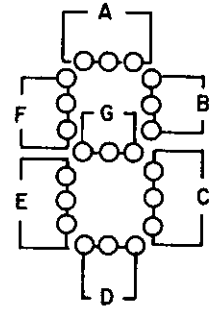
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Check the source to verify that this is the current version before use.

TRUTH TABLE														
CIRCUIT CONFIGURATION 3														
INPUT 2/							DISPLAY	SEGMENT LIT 2/						
A	B	C	D	E	F	G		A	B	C	D	E	F	G
○	○	○	○	○	○		0	△	△	△	△	△	△	
	○	○					1		△	△				
○	○		○	○		○	2	△	△		△	△		△
○	○	○	○			○	3	△	△	△	△			△
	○	○			○	○	4		△	△			△	△
○		○	○		○	○	5	△		△	△		△	△
○		○	○	○	○	○	6	△		△	△	△	△	△
○	○	○					7	△	△	△				
○	○	○	○	○	○	○	8	△	△	△	△	△	△	△
○	○	○	○		○	○	9	△	△	△	△		△	△
○	○	○		○	○	○	A	△	△	△		△	△	△
		○	○	○	○	○	b			△	△	△	△	△
○			○	○	○		C	△			△	△	△	
	○	○	○	○		○	d		△	△	△	△		△
○			○	○	○	○	E	△			△	△	△	△
○				○	○	○	F	△				△	△	△

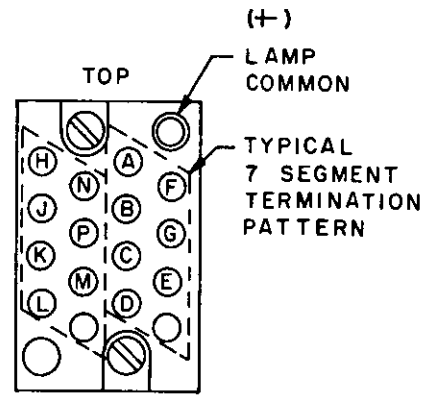
2/ Typical 7 segment pattern repeats as A thru G, H thru P, and R thru X



FRONT VIEW - 7 SEGMENT
(BAR MATRIX)



FRONT VIEW - 7 SEGMENT
(DOT MATRIX)

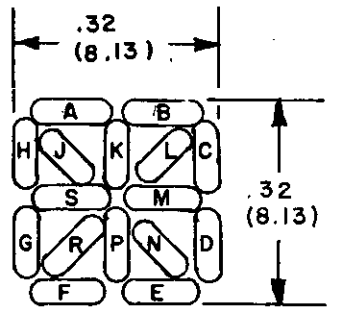


REAR VIEW - TERMINATION

TABLE III. Truth table - seven segment direct wired numeric alpha display.

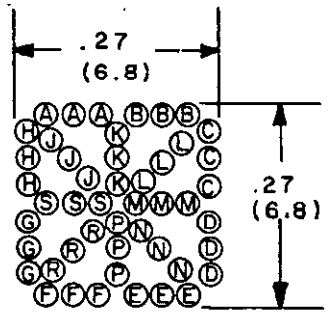
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:00Z
Check the source to verify that this is the current version before use.

TRUTH TABLE CIRCUIT CONFIGURATION 4 INPUT / SEGMENT LIT																
DISPLAY	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S
A	△	△	△	△			△	△				△				△
B	△	△	△	△	△	△				△		△		△		
C	△	△			△	△	△	△								
D	△	△	△	△	△	△				△				△		
E	△	△			△	△	△	△				△				△
F	△	△					△	△				△				△
G	△	△		△	△	△	△	△				△				
H			△	△			△	△				△				△
I	△	△			△	△				△				△		
J			△	△	△	△	△									
K							△	△			△		△			△
L					△	△	△	△								
M			△	△			△	△	△		△					
N			△	△			△	△	△				△			
O	△	△	△	△	△	△	△	△								
P	△	△	△				△	△				△				△
Q	△	△	△	△	△	△	△	△					△			
R	△	△	△				△	△				△	△			△
S	△	△		△	△	△		△				△				△
T	△	△								△				△		
U			△	△	△	△	△	△								
V							△	△			△				△	
W			△	△			△	△					△		△	
X									△		△		△		△	
Y									△		△			△		
Z	△	△			△	△					△				△	



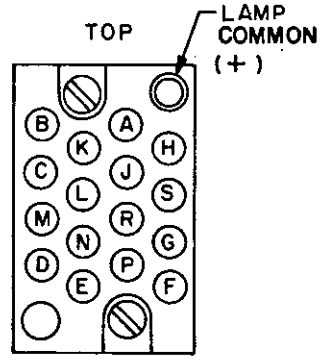
FRONT VIEW
16 SEGMENT DESIGNATIONS

(BAR MATRIX)



FRONT VIEW
16 SEGMENT DESIGNATIONS

(DOT MATRIX)

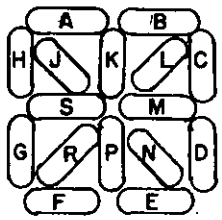


REAR VIEW
TERMINATIONS

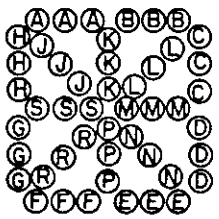
TABLE IV. Truth table - alpha/numeric direct wire - alphabetic characters.

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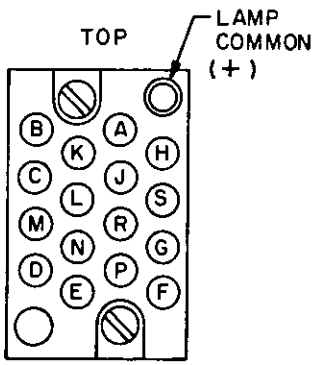
TRUTH TABLE CIRCUIT CONFIGURATION 5 INPUT / SEGMENT LIT																
DISPLAY	A	B	C	D	E	F	G	H	J	K	L	M	N	P	R	S
0		△	△	△	△					△				△		
1										△				△		
2		△	△		△							△		△		
3		△	△	△	△							△				
4			△	△						△		△				
5		△		△	△					△		△				
6		△		△	△					△		△		△		
7		△	△	△												
8		△	△	△	△					△		△		△		
9		△	△	△	△					△		△				
10		△	△	△	△		△	△		△				△		
11			△	△			△	△								
12		△	△		△		△	△				△		△		
13		△	△	△	△		△	△				△				
14			△	△			△	△		△		△				
15		△		△	△		△	△		△		△				
16		△		△	△		△	△		△		△		△		
17		△	△	△			△	△								
18		△	△	△	△		△	△		△		△		△		
19		△	△	△	△		△	△		△		△				
+										△		△		△		△
−												△				△
/											△				△	
\									△				△			
9	△	△		△	△	△		△		△		△		△		△
▷							△	△	△						△	
◁			△	△							△		△			



FRONT VIEW
16 SEGMENT DESIGNATIONS
(BAR MATRIX)



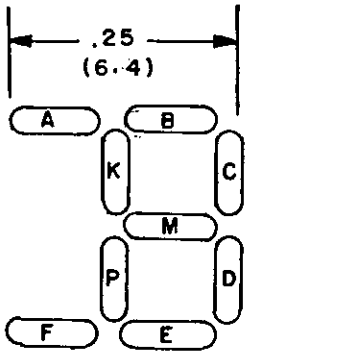
FRONT VIEW
16 SEGMENT DESIGNATIONS
(DOT MATRIX)



REAR VIEW
TERMINATIONS

TABLE V. Truth table - alpha/numeric direct wire - numeric characters/symbols.

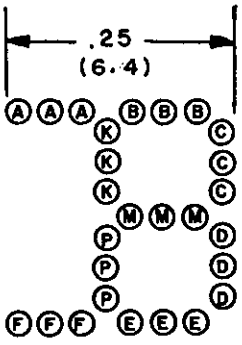
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FRONT VIEW - 9 SEGMENT

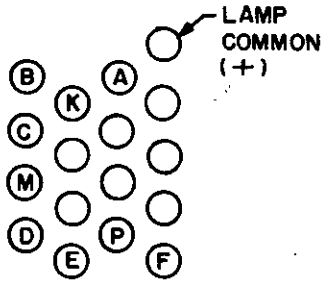
(BAR MATRIX)

TRUTH TABLE									
CIRCUIT CONFIGURATION 6									
	INPUT/SEGMENT LIT								
	A	B	C	D	E	F	K	M	P
0		△	△	△	△		△		△
1			△	△					
2		△	△		△			△	△
3		△	△	△	△			△	
4			△	△			△	△	
5		△		△	△		△	△	
6				△	△		△	△	△
7		△	△	△					
8		△	△	△	△		△	△	△
9		△	△	△			△	△	
A		△	△	△			△	△	△
B	△	△	△	△	△	△	△	△	△
C		△			△		△		△
D	△	△	△	△	△	△	△		△
E		△			△		△	△	△
F		△					△	△	△



FRONT VIEW - 9 SEGMENT

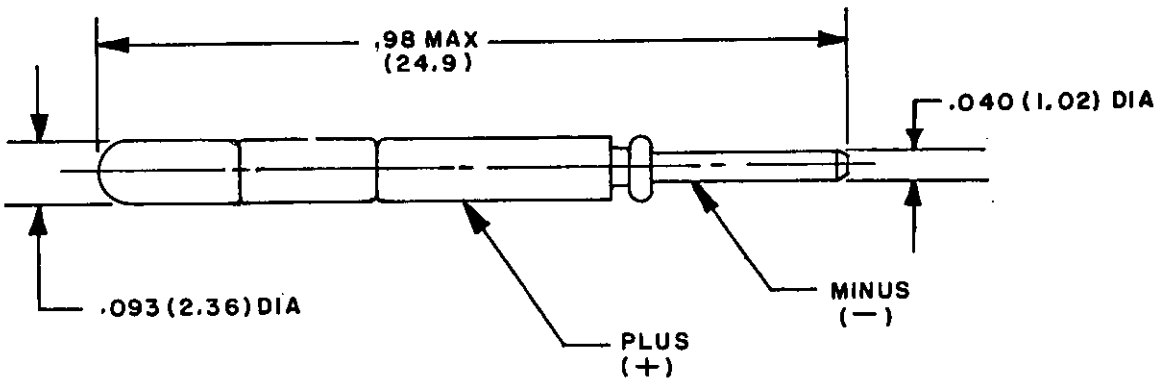
(DOT MATRIX)



REAR VIEW TERMINATION

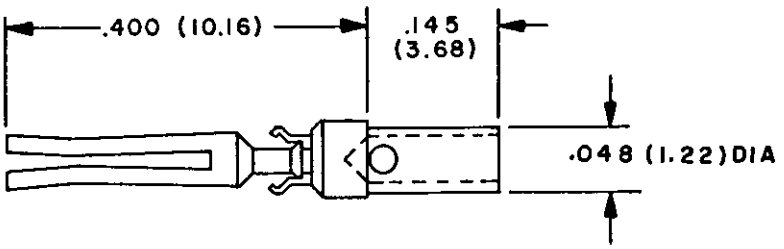
TABLE VI. Truth table - nine segment direct wired numeric alpha display.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:00Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions are in inches.
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3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional with envelope as shown.

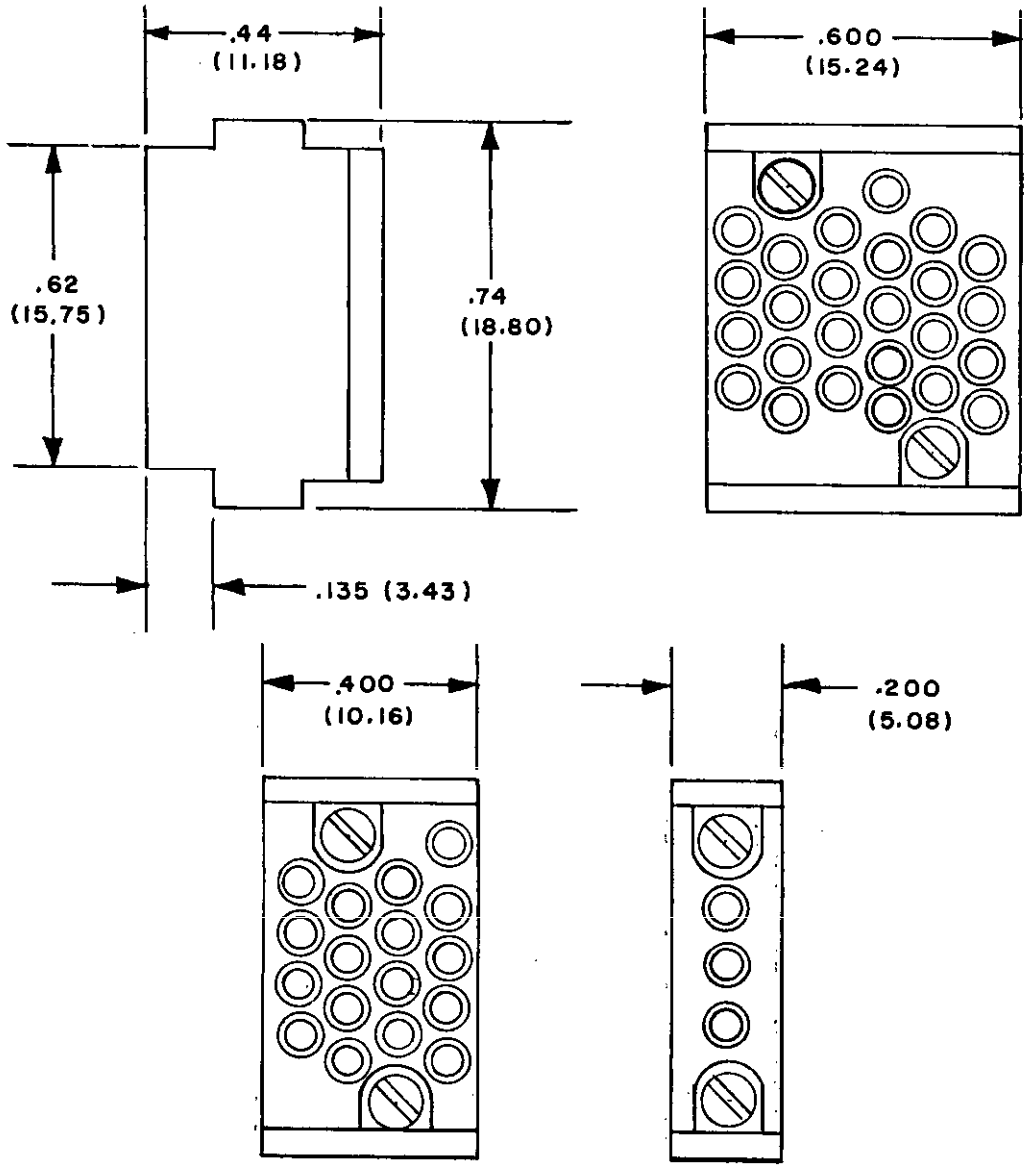
FIGURE 4. Lamp/diode assembly.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional with envelope as shown.

FIGURE 5. Terminal, solder type.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional with envelope as shown.

FIGURE 6. Connector blocks.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1 through 6, and tables I through VIII.

Weight: See table VII.

Light source: Lamp assembly includes JAN 1N4531 isolation diode and T-3/4 incandescent.

Storage and operating temperature: -55°C to +85°C.

Vibration grade: 2 installed.

Shock: Method I installed (50 g).

Chromaticity: Method I. See MIL-R-28803/5 for illuminated chromaticity limits.

Luminance: Average 300 foot-lamberts minimum with a supply voltage of 5 volts.

Dot to dot ratio:

Segment to segment: 3 to 1 maximum.

Digit to digit: 3 to 1 maximum.

Electrical:

Supply voltage: 5 volts maximum.

Supply current: (Each lamp) .025 ampere maximum.

Part number: The part number shall consist of the letter "M" followed by the basic number of the specification sheet and with one or two digit symbol indicating the type of module, connector block, terminal, or lamp.

M28803/6 A

Military designator and specification sheet no.

Code indicating type of module, connector block, terminal, or lamp (see table VIII).

Intended use: MIL-R-28803/6 is intended for use as a part of readout assemblies covered by MIL-R-28803/5.

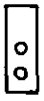
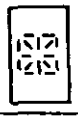
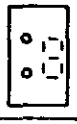
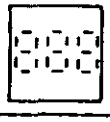
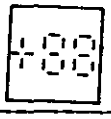
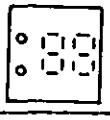
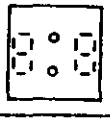
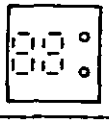
TABLE VII. Weight of modules (by code letter). 1/

Module code letter	Maximum weight (grams)	Module code letter	Maximum weight (grams)
A-A1	8.0	L-L1	14.0
B-B1	11.0	M-M1	11.0
C-C1	11.0	N	2.0
D-D1	11.0	P	6.0
E-E1	11.0	Q	8.0
F-F1	11.0	R	2.0
G-G1	14.0	S	0.2
H-H1	14.0	T-U-T1-U1	11.0
J-J1	14.0	V-V1	11.0
K-K1	14.0	W-W1	11.0

1/ For explanation of module code letter, see table VIII.

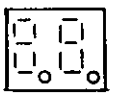
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TABLE VIII. Code letters for type of module assemblies connector blocks, terminals and lamp assemblies.

Bar matrix code letter	Dot matrix code letter	Display	Type of module
A1	A		Colon, degree or decimal
B1	B		16 segment alpha/numeric
C1	C		2 - 7 segment numeric
D1	D		4 segment sign and 7 segment numeric
E1	E		Colon and 7 segment numeric
F1	F		7 segment numeric and colon
G1	G		3 - 7 segment numeric
H1	H		1 - 4 segment and 2 - 7 segment numeric
J1	J		Colon and 2 - 7 segment numeric
K1	K		7 segment, colon and 7 segment numeric
L1	L		2 - 7 segment, numeric and colon
M1	M		1 - 9 segment alpha numeric

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TABLE VIII. Code letters for type of module assemblies connector blocks, terminals and lamp assemblies - Continued.

Bar matrix code letter	Dot matrix code letter	Display	Type of module
N	N	.200 wide	Connector block (figure 6) 1/
P	P	.400 wide	Connector block (figure 6) 1/
Q	Q	.600 wide	Connector block (figure 6) 1/
R	R	---	Terminals (figure 5) 10 to the bag
S	S	---	Lamp/diode assembly (figure 4)
T1	T		North-South
U1	U		East-West
V1	V		2 - 7 segment with decimal
W1	W		2 - 7 segment with 2 decimal

1/ For government acquisitions, connector blocks shall be furnished with the required number of terminals plus two extra.

Custodians:
Navy - EC
Air Force - 11

Review activities:
Navy - AS
DLA - ES, GS

User activities:
Army - AR, AT, AV, ME
Navy - MC, OS

Preparing activity:
Navy - EC

Agent:
DLA - ES

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